



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
Faculty of Computing

Project Task 2

(Preliminary Analysis)

Programme	: Bachelor of Computer Science (<i>Data Engineering</i>)
Subject Code	: SECR1213
Subject Name	: Network Communications
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Section : 12

Group : 12.3 – 4Fabs

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Question

1. What are the current pain points with the existing networking setup?

There are frequent connectivity issues during peak usage times, especially in high-density areas like lecture halls and labs and also some areas with lower or even no connection. We also experience slower speeds and intermittent dropouts during video conferences and virtual meetings, which can disrupt communication and collaboration.

2. What types of network devices will be required for the Cisco Network lab, video conferencing room and embedded lab?

The Cisco Network Lab will require appropriate networking equipment, such as routers, switches, and firewalls, to facilitate the teaching of Cisco-related concepts.[6] Similarly, the video conferencing room will necessitate high-quality audiovisual components, including cameras, microphones, speakers, and a reliable, high-bandwidth internet connection, to enable effective remote communication and collaboration. Lastly, the embedded lab will need a range of specialized hardware, including logic analyzer and Raspberry Pi kits, microcontrollers, digital oscilloscopes, and sensor modules, in addition to relevant software tools like MATLAB, LabVIEW, and IoT cloud platforms, to support the hands-on exploration and development of embedded systems.[7, 8]

3. What are the essential network and hardware requirements for an effective and high quality video conferencing room?

The hardware requirement for an effective and high-quality video conferencing room is having a high-resolution webcam. It is essential for clear video communication, as it captures subtle details like facial expressions and body language, enhancing the authenticity of virtual interactions. The sufficient network bandwidth is also critical to achieve the requirement. For example, a moderate-quality 720p video at 15 frames per second requires about 1 Mbps, while a high-quality 4K video at 30 frames per second demands approximately 4 Mbps. The amount of internet strength needed depends on how clear you want the video to be. Additionally, when multiple participants join a video call, higher bandwidth is necessary to maintain smooth connectivity. Even with strong Wi-Fi coverage, simultaneous usage can strain the network, the network must be capable of handling concurrent usage to maintain uninterrupted and high-quality video conferencing.

4. What network services will be offered to support teaching and learning activities?

Provide virtual learning platforms hosted on local servers or the cloud, secure access to

digital libraries, research databases, and online courses. Implement streaming services for live lectures and virtual labs to enhance interactive learning. For example, Cisco Education which securely connects educators, students, and administrators with transformative technologies that power inclusive learning for all[15].

5. How can the building's wireless network support high-density areas like the student lounge and labs?

To effectively support high-density areas, such as the student lounge and labs, the building's wireless network should utilize enterprise-grade access points equipped with multi-user, multiple-input, multiple-output technology and band steering capabilities. Strategically positioning the access points can optimize wireless coverage. Additionally, implementing Wi-Fi 6 access points can facilitate simultaneous connections for numerous devices in each floor.[10] Further, segregating the wireless network into separate service set identifiers for students, staff, and guests, while imposing bandwidth restrictions for non-critical users, can help manage network usage efficiently.

6. What type of network architecture (e.g., star, mesh, bus, or hybrid) would best suit the faculty's needs, considering factors like scalability, reliability, performance, budget, and ease of maintenance?

A hybrid network architecture that combines star and mesh topology is the most suitable design for the new faculty. On the first floor, star topology will be used to connect all devices such as printers, desktops and wireless access points to a central switch to ensure the effectiveness and localized connectivity. Otherwise, mesh topology will be used to link Floor 2's switches to Floor 1's main switch using a high-speed backbone such as fiber-optic or high-capacity Ethernet. This design balances scalability, performance, and cost, enabling easy additions of devices and robust communication between floors.

7. What is the best method of connecting devices to the network (e.g., wired, wireless, or a combination of both)?

The best method for connecting devices to the network is a combination of both wired and wireless connection depending on the type of device and usage. For wired networks, devices are physically connected by using cables to a network switch or router.[20] Wired networks are suitable for devices such as printers, desktop as they are able to provide a stable and high-speed connection with low latency. However, wireless connection using electromagnetic waves to transmit data from end-to-end. Devices like laptops, tablets and smartphones are more suitable to use wireless connection as it is more flexible without

using physical cable and ease of installation[20]. Otherwise, Wireless Access Points (APs) are strategically placed throughout the building to provide seamless coverage, allowing users to connect to the network with more stability.

8. What kind of wireless network setup is required to provide seamless connectivity across the building?

To provide seamless wireless connectivity across the faculty building, deploy multiple access points (APs) to ensure consistent coverage in all areas. Use wireless repeaters or extenders to eliminate dead zones in distant or obstructed spaces. Incorporate network switches and controllers for centralized management of APs, streamlining configuration, and improving network performance. Additionally, consider using directional or omnidirectional antennas based on the coverage requirements and ensure all devices are equipped with network interface cards (NICs) for compatibility with the wireless network[12].

9. What type of network cable is the most appropriate for connecting the computers, and what are the reasons for choosing it?

Cat6 Ethernet cables are the most appropriate type of cabling for the network setup. Cat6 cables support speeds up to 1Gbps, which is sufficient for typical educational applications and ideal for labs with the need of high speed connectivity[16]. Additionally, Cat6 cables have a good balance between performance and cost as they are widely available and easy to source. Cat6 cables are compatible with existing networking equipment such as routers and switches that support Gigabit Ethernet, ensuring that the faculty's network infrastructure work efficiently without requiring an upgrade[16].

10. How can network efficiency and low latency be achieved?

Low latency and network efficiency require the use of several techniques. A network design that is well-structured, like a core-distribution-access topology, guarantees effective traffic flow by lowering congestion and the distance that data must travel. Important applications like video conferencing may function without interruption thanks to network segmentation using VLANs, which helps isolate various traffic types.[2] By prioritising time-sensitive traffic, Quality of Service (QoS) implementation guarantees minimal latency for applications with high demand.[1] To maximise speed and reduce delays, high-performance gear is necessary. Examples of this hardware include fiber-optic cabling for inter-floor connections, low-latency access points, and gigabit or 10-gigabit Ethernet switches. The network is further optimised by traffic prioritization including load balancing, traffic shaping, and redundancy, which distribute traffic uniformly and offer backup routes to avoid congestion.[1]

11. What bandwidth and internet connectivity are required to support the faculty's operations?

A minimum bandwidth of 1 Gbps is recommended, with provisions to scale to 10 Gbps in the future[11]. Bandwidth is critical to the success of hardware initiatives that attempt to provide ubiquitous access to learning, such as Bring Your Own Device (BYOD) and 1:1 laptop and tablet programs. The amount of bandwidth needed is obviously driven by the types of networked activities in which the user is engaged such as downloading content, online professional developments and online testing.

12. How can we ensure that the wireless network covers the entire faculty building effectively?

To ensure that the coverage wireless network is able to support the area of the faculty, a thorough evaluation of signal strength across all areas, including floors, common spaces and specific rooms is essential[19]. This evaluation involves determining the quantity of current Access Points (APs) and their effective range to establish the extent of network coverage [19]. Detect and scrutinize dead zones, areas with weak or non-existent signal strength [19], to pinpoint areas in need of improved coverage. Considering the characteristics of the wireless network, including factors like Wi-Fi channels, network authentication, and security protocols, make necessary adjustments to optimize coverage and ensure a dependable connection. Based on the findings and analysis, identify suitable new locations for the placement of APs to enhance coverage and address the previously identified dead zones. Carefully position the new APs to ensure high-quality speed and coverage precisely where it's needed.

13. How should the building's network be secured against external and internal threats?

The Cybersecurity and Infrastructure Security Agency (CISA) encourages users and network administrators to implement the segment and segregate networks and functions to better secure their network infrastructure. Proper network segmentation is an effective security mechanism to prevent an intruder from propagating exploits or laterally moving around an internal network. A securely segregated network can contain malicious occurrences, reducing the impact from intruders in the event that they have gained a foothold somewhere inside the network[13]. Moreover, implementing a Virtual Private Network (VPN) is essential[14]. A VPN encrypts data transmissions and masks users' IP addresses, providing secure access to the network, especially for remote users or those connecting via public Wi-Fi. This helps protect sensitive information from unauthorized access and cyber threats.

14. What disaster recovery plan is needed for network downtime?

Planning for disaster recovery is essential to ensure an organization can maintain operations during unexpected events such as natural disasters, cyberattacks, or other emergencies. Regular data backups can ensure the important files and settings are safe and can be recovered easily. Implementing a backup network system, like additional routers and internet connections can keep the network running if the main system fails[17]. The real-time monitoring tools play a crucial role in detecting issues early by sending alerts, so issues can be addressed quickly[17]. Furthermore, Uninterruptible Power Supplies(UPS) can be used to protect devices from power outages as it can give more time to fix the issue without losing connectivity[18]. These measures collectively strengthen the organization's resilience and minimize the impact of disruptions.

15. What monitoring tools are recommended for detecting and managing network issues?

Several monitoring technologies can be used to provide real-time insights, identify bottlenecks, and guarantee optimal performance in order to detect and address network issues efficiently. With comprehensive data on bandwidth utilisation, latency, and device condition, SolarWinds Network Performance Monitor is a well-liked tool for keeping an eye on the functionality and health of network devices. It can detect and solve network issues before they cause downtime.[3] With customisable alerts to inform administrators of any problems, PRTG Network Monitor provides a complete solution for tracking network traffic, server performance, and device health. Another open-source tool for network and infrastructure monitoring is Nagios, which can measure a number of metrics like response times, availability, and uptime.[3] In order to solve problems like network congestion or security vulnerabilities, network administrators can collect and examine packets using Wireshark, which allow us to capture and analyze network traffic. [4]

Feasibility Assessment

Feasibility assessments are critical in determining whether a project is realistic and worth pursuing. After the Question and answer part, we can now evaluate and discuss the feasibility of our project. These are the four feasibility we are going to determine and evaluate:

1. Technical Feasibility

Based on the information gathered, the project appears to be technically feasible. The proposed network architecture features a fiber-optic backbone and Cat6 cabling to support high-speed connectivity and scalability for faculty. We also select enterprise-grade access points with Wi-Fi 6 to ensure coverage in high-density areas like labs and lounges to meet future demands. The integration of devices such as Raspberry Pis, microcontrollers, and Cisco equipment meets the labs' technical and educational needs. Lastly, the adoption of modern technologies like VLANs for traffic segmentation, Quality of Service (QoS) for latency-sensitive applications, and monitoring tools like SolarWinds and Nagios enhance network reliability and performance.

2. Economic Feasibility

It is essential to evaluate the costs and benefits of the proposed investment to assess the economic feasibility of the project. The initial costs include purchasing equipment, software installation and software license fees while operational costs include maintenance, IT management staffing and power consumption. However, the project offers significant economic advantages, such as reduced downtime, increase productivity, and long-term savings from scalable and future-proof technology like Wi-Fi 6 and Cat6 Ethernet cabling. A cost-benefit analysis which compare the costs with the benefits it will provide will helps in ensuring the project stays within budget while achieving the faculty's goals effectively.

3. Operational Feasibility

The operational feasibility of the project appears promising. By using a hybrid network architecture which combines star and mesh topologies helps to ensure the connectivity and seamless communication between floors. Otherwise, high-performance hardware such as managed switches, routers and Wi-Fi 6 access points support efficient wired and wireless connections. Moreover, for solving traffic segregation, we have included VLANs and Quality of Service(QoS) to keep the lower latency while using video conferencing and hybrid labs. Monitoring tools like SolarWinds enable proactive issue management. While strong security standards protect data, but disaster recovery methods, such as backups and redundant systems can improve resilience.

4. Security Consideration

To secure the faculty building's network, segmentation and segregation of networks are essential to contain threats and limit intruder movement within the network.

Implementing Virtual Private Networks (VPNs) ensures encrypted communication for remote access and public wi-Fi usage. A robust disaster recovery plan is necessary such as redundant systems and regular backups, making the network secure and reliable for educational operations.

MEETING MINUTES #4

DATE/TIME	22 Nov 2024 10am		
LOCATION	Online Webex Meeting		
AGENDA	1. Discuss about all the task from task 2 2. Suggest possible questions for task 2 3. Select the most related questions from all suggestion 4. Task distribution for each person		
Meeting MC	Woo Cheng Shuan		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Poh Lok Yee	1000	-	
Woo Cheng Shuan	1000	-	
Chau Ying Jia	1000	-	
Sabrina Heng Wei Qi	1000	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Discussion for task 2	All members read and understand requirements from task2.	All members
2	Suggest possible questions	Each member contributes 5-6 questions regarding the case study.	All members

3	Select the most related questions	<i>Sabrina lists out the questions from each member in the Word document to easier members to read in overall.</i> <i>All members read and understand the questions provided by other members.</i> <i>All members discussed the questions one by one and selected 20 most related questions that can be used in task 2.</i>	All members
4	Task distribution	<i>Lok Yee distributed the task to all members equally.</i> <i>Each member gets 5 questions.</i> <i>All members had to do research to find a solution for those questions with listing out the references.</i>	Lok Yee
5	Next Meeting	<i>26 November – All given tasks should be 95% completed</i>	All members
6	Meeting ended	<i>1300</i>	All members

MEETING MINUTES #5

DATE/TIME	26 Nov 2024 10am		
LOCATION	Online Webex Meeting		
AGENDA	1. Review the answers for each question 2. Finalize the most suitable questions to be used 3. Discuss on the feasibility assessment		
Meeting MC	Poh Lok Yee		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Poh Lok Yee	1000	-	
Woo Cheng Shuan	1000	-	
Chau Ying Jia	1000	-	
Sabrina Heng Wei Qi	1000	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Explain answer for each question	All members explained their responsive questions and answers one by one.	All members
2	Review the answer	All members review the questions and answer carefully	All members
3	Finalize the most related	Sabrina suggested dropping the questions with answers	All members

	questions and answers	<i>that are not from any reference.</i> <i>Ying Jia suggested dropping the questions that are quite related.</i> <i>Cheng Shuan said that some answers are confusing.</i> <i>Lok Yee highlighted the question that can be used.</i> <i>Lok Yee finalizes 15 questions and answers.</i> <i>All members agree with the finalized list.</i>	
4	Discuss on the feasibility assessment	<i>Lok Yee explained the feasibility assessment to all members.</i> <i>Ying Jia suggested covering several aspects in this assessment.</i>	Lok Yee and Ying Jia
5	Meeting ended	<i>At 1230, the meeting ended after all discussions were done.</i>	All members

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